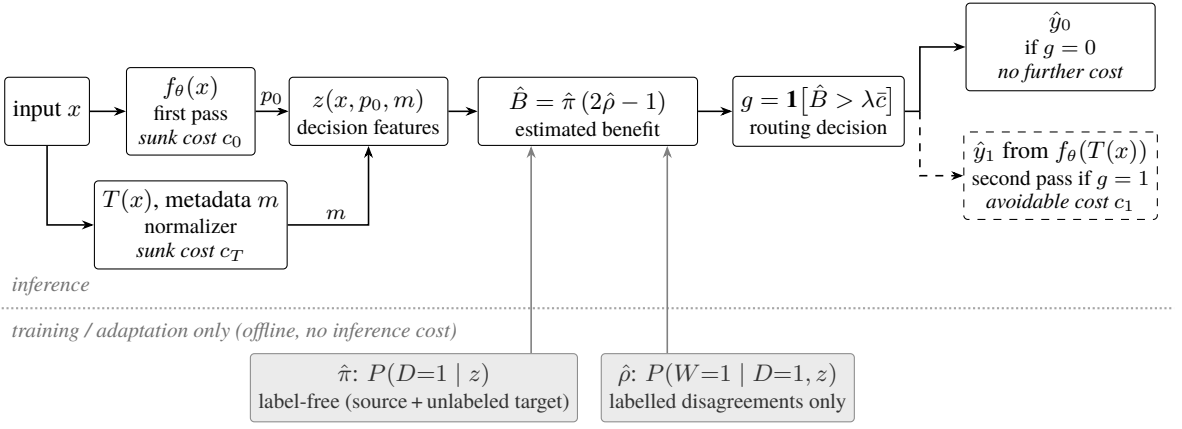

input x
$f_\theta(x)$
first pass
sunk cost c_0
p_0
$z(x, p_0, m)$
decision features
$\hat{B} = \hat{\pi}\,(2\hat{\rho} - 1)$
estimated benefit
$g = \mathbf{1}\big[\hat{B} > \lambda\bar{c}\big]$
routing decision
$\hat{y}_0$
if $g = 0$
no further cost
$\hat{y}_1$ from $f_\theta(T(x))$
second pass if $g = 1$
avoidable cost c_1
$T(x)$, metadata m
normalizer
sunk cost c_T
m
inference
training / adaptation only (offline, no inference cost)
$\hat{\pi}$: $P(D{=}1 \mid z)$
label-free (source + unlabeled target)
$\hat{\rho}$: $P(W{=}1 \mid D{=}1, z)$
labelled disagreements only